Strong generalization on constants

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs thm:WeakGen, lem:ChangeBdVar (both this batch); the chain WeakGen -> ChangeBdVar -> StrongGen is delivered as three paired cards.

Editorial note. The label thm:StrongGen names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Weak generalization on constants. Same unit, \ollabel{thm:WeakGen} (source line 1326, this batch). First step: produces the variable y and the proof of forall y subst(A,c,y).
- Change of bound variable. Same unit, \ollabel{lem:ChangeBdVar} (source line 1373, this batch). Second step: the biconditional forall x subst(A,x,c) iff forall y subst(A,y,c).

Theorem 0.1 (Strong Generalization on Constants)**.** *If $\Gamma \vdash A$, the constant c does not occur in Γ, and x is not free in A but it is free for c in A, then $\Gamma \vdash \forall x\, A[x/c]$.*

Proof. Since $\Gamma \vdash A$ and c is not in Γ, then by Weak Generalization there is variable y not in A such that $\Gamma \vdash \forall y\, A[y/c]$. Since y is not in A (at all), it is not free in A and it is also free for c in A; if moreover (by hypothesis) x is not free in A and free for c in A, then the requirements for a change of bound variable are met, so $\vdash \forall x\, A[x/c] \equiv \forall y\, A[y/c]$, whence $\Gamma \vdash \forall x\, A[x/c]$. □